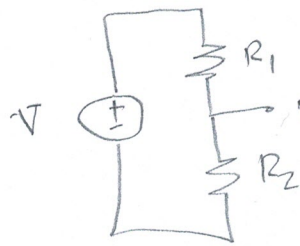


2.42

a)



$$V = 50V$$

$$R_1 = 15k\Omega$$

$$i = (R_1 + R_2)^{-1} V$$

SPÄNNINGSFALLET ÖVER R_2
SKALL VARA V_U

$$V_U = R_2 i = \frac{V R_2}{R_1 + R_2}$$

$$V_U (R_1 + R_2) = V R_2$$

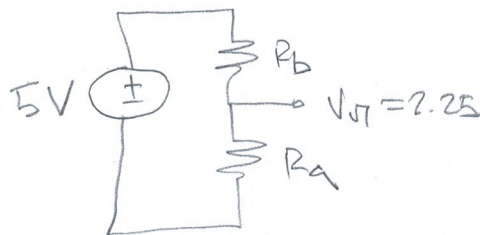
$$R_2 = \frac{V_U R_1}{V - V_U} =$$

$$= \frac{20 \cdot 15 \cdot 10^3}{50 - 20} = 10k\Omega$$

$$\text{EFFEKT } P = V_U i = V_U \frac{V_U}{R_2} = \frac{V_U^2}{R_2} =$$

$$= \frac{20^2}{10 \cdot 10^3} = 2 \cdot 10^{-2} = 40 \text{ mW}$$

b) På SUMMA SÄTT MED $V_U = 2.25V$
På SUMMA SOM OAV, $10k\Omega$



$$R_b = \frac{V - V_U}{V_U} R_a =$$

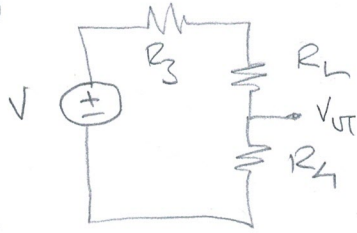
$$= \frac{5 - 2.25}{2.25} 10 \cdot 10^3 =$$

$$= 12222\Omega$$

$$\text{OCH EFFEKTEN } P_{V_{R_1}} = V_{R_1}^2 \frac{1}{R_1} =$$

$$= \frac{(5 - 2.25)^2}{12222} = 0.619 \text{ mW}$$

c)



$$V = 110 \text{ V}$$

$$R_3 = 1 \text{ k}\Omega$$

$$R_4 = 2.7 \text{ k}\Omega$$

$$V_{U7} = 28.3 \text{ V}$$

Stromen BUIR

$$i = \frac{V}{R_3 + R_L + R_4}$$

$$\text{oder } V_{U7} = R_4 i =$$

$$= \frac{R_4 V}{R_3 + R_L + R_4}$$

$$\Rightarrow V_{U7} (R_3 + R_L + R_4) = R_4 V$$

Lös für \$R_L\$

$$R_L = R_4 \frac{V}{V_{U7}} - R_3 - R_4 =$$

$$= 2.7 \cdot 10^3 \frac{110}{28.3} - 1 \cdot 10^3 - 2.7 \cdot 10^3 =$$

$$\approx \underline{\underline{6794.7 \Omega}}$$

SVAR: a) \$R_2 = 10 \text{ k}\Omega\$

b) \$R_1 = 12222 \Omega\$

c) \$R_L = 6795 \Omega\$